

Key Concepts in Calculus

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1 Acceleration, Position, and Velocity

Let's start with acceleration. Imagine you hit the gas in your stopped car as hard as you can. We can represent this constant acceleration as:

$$a(t) = a_0 \quad (1)$$

As a result, as illustrated in Figure 1 the velocity increases linearly over time:

$$v(t) = a_0 t \quad (2)$$

Initially, the position $y(t)$ does not change much because the velocity is low. However, as velocity grows, your position changes rapidly. This means $y(t)$ is.

$$y(t) = \frac{1}{2} a_0 t^2 \quad (3)$$

We can see this visually in Figure 2, where the slope is initially flat (“small slope”) and grows increasingly steep (“high slope”).

Let's confirm that all of these formulas are true by working backwards. As illustrated in Figure 3, velocity is the change in position (Δy) divided by the change in time (Δt).

$$v(t) \approx \frac{\Delta y}{\Delta t} = \frac{y(t + \Delta t) - y(t)}{\Delta t} \quad (4)$$

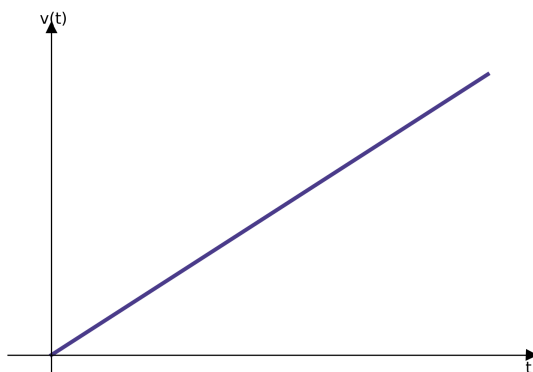


Figure 1: Velocity $v(t)$ as a function of time under constant acceleration $a(t) = a_0$.

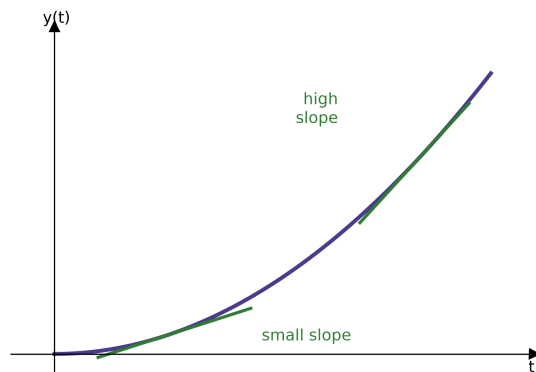


Figure 2: Position $y(t)$ as a function of time.

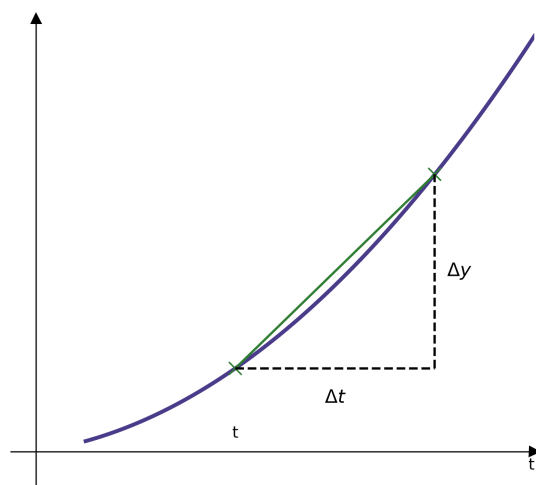


Figure 3: The approximate velocity $\Delta y / \Delta t$ at a time t on a graph of the position $y(t)$.

Plugging in our formula for $y(t)$:

$$\begin{aligned}
\frac{\Delta y}{\Delta t} &= \frac{\frac{1}{2}a_0(t + \Delta t)^2 - \frac{1}{2}a_0t^2}{\Delta t} \\
&= \frac{\frac{1}{2}a_0(t + \Delta t)(t + \Delta t) - \frac{1}{2}a_0t^2}{\Delta t} \\
&= \frac{\frac{1}{2}a_0[t^2 + 2t\Delta t + \Delta t^2] - \frac{1}{2}a_0t^2}{\Delta t} \\
&= \frac{\frac{1}{2}a_0t^2 + a_0t\Delta t + \frac{1}{2}a_0\Delta t^2 - \frac{1}{2}a_0t^2}{\Delta t} \\
&= \frac{a_0t\Delta t + \frac{1}{2}a_0\Delta t^2}{\Delta t} \\
&= a_0t + \frac{1}{2}a_0\Delta t
\end{aligned} \tag{5}$$

To make this last formula exact, we consider finer and finer timesteps until Δt goes to zero. Plugging in zero for Δt yields

$$v(t) = \frac{dy}{dt} = a_0t \tag{6}$$

This yields the velocity formula we assumed in Equation 2.

Applying the same procedure to find the acceleration from this velocity formula yields

$$\begin{aligned}
a(t) &\approx \frac{\Delta v}{\Delta t} = \frac{v(t + \Delta t) - v(t)}{\Delta t} \\
&= \frac{a_0(t + \Delta t) - a_0t}{\Delta t} \\
&= \frac{a_0t + a_0\Delta t - a_0t}{\Delta t} = a_0
\end{aligned} \tag{7}$$

This again yields the acceleration equation we assumed in Equation 1. In this case, because the velocity was strictly linear, our approximation turned out to be exact.

2 Summary of Calculus Operations

In general:

- If you know position $y(t)$, you can take the derivative to find velocity and acceleration:

$$y(t) \xrightarrow{\frac{d}{dt}} v(t) \xrightarrow{\frac{d}{dt}} a(t)$$

- If you know acceleration $a(t)$, you can take the integral to find velocity and position (introducing constants of integration $+C$):

$$v(t) = \int a(t) dt + v_0 \tag{8}$$

$$y(t) = \int v(t) dt + y_0 \tag{9}$$

Table 1: Common Derivatives and Integrals

Function $f(x)$	Derivative $\frac{d}{dx}f(x)$	Integral $\int f(x) dx$
x^n	nx^{n-1}	$\frac{x^{n+1}}{n+1} + C \quad (n \neq -1)$
$\sin(x)$	$\cos(x)$	$-\cos(x) + C$
$\cos(x)$	$-\sin(x)$	$\sin(x) + C$
e^x	e^x	$e^x + C$
$\ln(x)$	$\frac{1}{x}$	$x \ln(x) - x + C$
$\frac{1}{x}$	$-\frac{1}{x^2}$	$\ln x + C$

Constant Multiple: $\frac{d}{dx}[c \cdot u(x)] = c \frac{du}{dx}$

Sum Rule: $\frac{d}{dx}[u(x) + v(x)] = \frac{du}{dx} + \frac{dv}{dx}$

Product Rule: $\frac{d}{dx}[u(x) \cdot v(x)] = \frac{du}{dx}v(x) + u(x)\frac{dv}{dx}$

Quotient Rule: $\frac{d}{dx} \left[\frac{u(x)}{v(x)} \right] = \frac{v(x)\frac{du}{dx} - u(x)\frac{dv}{dx}}{[v(x)]^2}$

Chain Rule: $\frac{d}{dx}[u(v(x))] = \frac{du}{dv} \cdot \frac{dv}{dx}$

Integration by Parts: $\int u dv = uv - \int v du$

3 Motivating Key Calculus Results

Here are some motivations (maybe not full mathematical “proofs” in every case) of the results in Table 1. There are of course full mathematical proofs of all of these results available in textbooks and the literature, but the intent here is to motivate the results so we can apply them to real problems.

3.1 Sine and Cosine

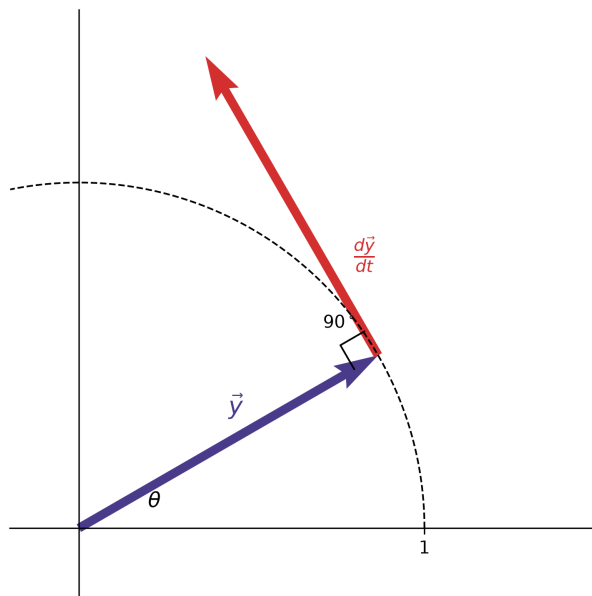


Figure 4: A rotating vector of length 1.

Let’s draw a vector $\vec{y}$ that has a length of 1 and is rotating about the center counterclockwise. At a certain moment it is at an angle θ . This is drawn in Figure 4. The coordinates of this vector are

$$\vec{y}(\theta) = (\cos \theta, \sin \theta). \quad (10)$$

The tip of the vector is moving at a 90° angle relative to the position vector. Drawing another vector $\frac{d\vec{y}}{dt}$ at 90° but keeping the same length as the original means swapping the two coordinates, and putting in a minus sign, yielding

$$\frac{d\vec{y}}{dt} = (-\sin \theta, \cos \theta). \quad (11)$$

Since this velocity is the derivative of the position, that means that

$$\frac{d}{dx} \cos(x) = -\sin(x) \quad (12)$$

$$\frac{d}{dx} \sin(x) = \cos(x), \quad (13)$$

just as is written in Table 1.

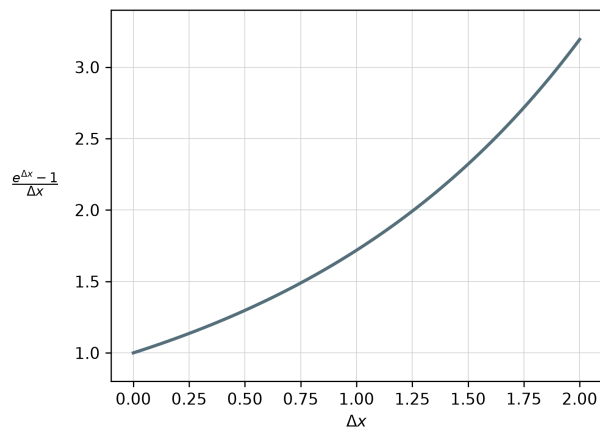


Figure 5: Plot of $\frac{e^{\Delta x}-1}{\Delta x}$ approaching 1 as Δx goes to zero.

3.2 Exponential

For $y = e^x$, we set up the definition of the approximate derivative:

$$\frac{\Delta y}{\Delta x} = \frac{e^{x+\Delta x} - e^x}{\Delta x} = \frac{e^x e^{\Delta x} - e^x}{\Delta x} = e^x \left(\frac{e^{\Delta x} - 1}{\Delta x} \right). \quad (14)$$

Let's examine the behavior of the term $\frac{e^{\Delta x}-1}{\Delta x}$ as Δx goes to zero by plotting it as shown in Figure 5. This shows that when Δx goes to zero, the term $\frac{e^{\Delta x}-1}{\Delta x}$ goes to 1. Therefore, putting this result into Equation 14 yields

$$\frac{d}{dx} e^x = e^x \quad (15)$$

which is exactly the result in Table 1.

3.3 Logarithm

Using this knowledge of the derivative of e^x , we can find the derivative of $y = \ln(x)$. First we rewrite this same equation for y using the relationship between logarithms and exponentials as

$$e^y = x. \quad (16)$$

Differentiating both sides with respect to y yields

$$\begin{aligned} \frac{d}{dy} e^y &= \frac{dx}{dy} \\ e^y &= \frac{dx}{dy} \\ x &= \frac{dx}{dy}. \end{aligned} \quad (17)$$

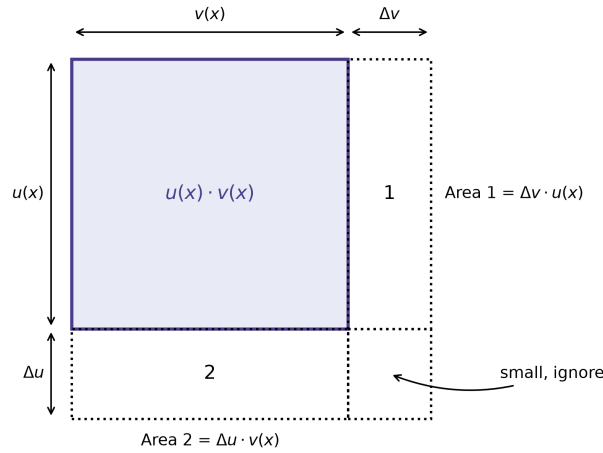


Figure 6: Geometric interpretation of the product rule using area changes.

Taking the reciprocal of both sides,

$$\begin{aligned} \frac{1}{x} &= \frac{dy}{dx} \\ \frac{1}{x} &= \frac{d}{dx} \ln(x), \end{aligned} \tag{18}$$

which is exactly what is written in Table 1.

3.4 Chain Rule

For the derivative of a composition of functions $y = u(v(x))$, we can think of dimensional analysis (i.e. unit analysis):

$$\frac{dy}{dx} = \frac{du}{dv} \cdot \frac{dv}{dx} \tag{19}$$

In some sense the dv “cancels out” yielding the desired result.

3.5 Product Rule

To find the derivative of a product $y = u(x) \cdot v(x)$, let’s look at the change in area of a rectangle with sides $u(x)$ and $v(x)$ when x increases by Δx . This is illustrated in Figure 6.

The total new area is $y(x + \Delta x) = u(x + \Delta x) \cdot v(x + \Delta x)$. The change in area Δy is composed of two components (ignoring the small corner):

- **Area 1:** $u(x) \cdot \Delta v$
- **Area 2:** $\Delta u \cdot v(x)$

Dividing by Δx :

$$\frac{\Delta y}{\Delta x} = \frac{\Delta u}{\Delta x} v(x) + u(x) \frac{\Delta v}{\Delta x} \quad (20)$$

As Δx goes to zero, this yields the product rule formula:

$$\frac{d}{dx}[u(x) \cdot v(x)] = \frac{du}{dx} v(x) + u(x) \frac{dv}{dx} \quad (21)$$

To memorize this, the classic sentence to say repeatedly to yourself is “one prime two, plus one two prime” where “prime” is short for d/dx . It’s not silly if it works...

3.6 Quotient Rule

We can combine the product rule, chain rule, and our knowledge of the derivative of $1/x$ to yield the quotient rule.

To find the derivative of a quotient

$$y = \frac{u(x)}{v(x)}, \quad (22)$$

we first rewrite this as a product

$$y = u(x) \frac{1}{v(x)}. \quad (23)$$

We then apply the product rule on $u(x)$ and $1/v(x)$ to get

$$\frac{dy}{dx} = u(x) \frac{d}{dx} \left[\frac{1}{v(x)} \right] + \frac{1}{v(x)} \frac{du}{dx}. \quad (24)$$

We can use the rule from Table 1 that gives the derivative of x^n by plugging in $n = -1$ to get that the derivative of $1/x$ is $-1/x^2$. Using this result together with the chain rule yields

$$\frac{d}{dx} \left[\frac{1}{v(x)} \right] = -\frac{1}{[v(x)]^2} \frac{dv}{dx}. \quad (25)$$

Substituting this into the product-rule expression yields

$$\frac{dy}{dx} = -\frac{u(x)}{[v(x)]^2} \frac{dv}{dx} + \frac{1}{v(x)} \frac{du}{dx} \quad (26)$$

Rewriting into the standard form, this yields the quotient rule:

$$\frac{d}{dx} \left[\frac{u(x)}{v(x)} \right] = \frac{v(x) \frac{du}{dx} - u(x) \frac{dv}{dx}}{[v(x)]^2} \quad (27)$$

To memorize this, the classic poem to say repeatedly to yourself is “low d high, less high d low, draw the line and down below, denominator squared will go” where “d” is meant to be short for d/dx . It’s not silly if it works...

3.7 Integration By Parts

Left as an exercise to the reader! Armed with this set of key calculus concepts, it's a great time to solve some problems, go back to a calculus textbook or other reference, and do great research!

AI Use Statement

The writing in this document was prepared entirely without AI. The concept for the derivation of the derivatives of the sine and cosine without the use of Taylor Series (i.e. the way the author learned it), as well as the concept for the graphical illustration for the product rule, were the result of interaction with generative AI. Both approaches have literature citations^{1 2}.

The concepts for all of the figures in this document was created without the use of AI. Generative AI used hand drawings of the figures to produce python code that produced the final figure images.

All interactions with generative AI were with the Google Gemini 3.5 Thinking model.

¹C. S. Ogilvy, Derivatives of sin and cos , American Mathematical Monthly 67 (1960) 673.

²Strang, G. (1991). Calculus. Wellesley-Cambridge Press.